

# NFR TESTING

Version: 3.0

Last Updated: 28 September 2026



Prepared by



# Non-functional Requirements

All targets are quantified using an observability platform of choice: Spring Boot Actuator (Micrometer) for application metrics, Prometheus and Grafana for collection and dashboards, UptimeRobot for uptime monitoring, and k6 for load generation.

## Performance

The system should respond to user actions within 1-2 seconds under normal load, with appropriate UI feedback (loading indicators, disabled controls while a request is in flight).

### *Quantification:*

- Normal load is 25 concurrent users.
- 95% of API requests complete within 2 seconds, with an error rate below 1%.

### *Measurement:*

k6 for load testing, Spring Boot Actuator and Prometheus for application metrics, Grafana for dashboards.

### *Example test:*

25 virtual users load projects, tasks and timesheets for 3 minutes. The test passes if p(95) is 2 seconds or less and fewer than 1% of requests fail.

## Scalability

The system must scale to support at least 100 concurrent users without performance degradation.

### *Quantification:*

At 100 concurrent users, 95% of API requests complete within 2 seconds, with an error rate below 1%.

### **Measurement:**

k6 for load testing, with Prometheus and Grafana tracking CPU, memory and database connection pool usage.

### **Example test:**

100 virtual users run the same workload for 3 minutes. The test passes if p(95) is 2 seconds or less and fewer than 1% of requests fail.

## **Availability**

Core services must be available 99.9% of the time, excluding scheduled maintenance.

### **Quantification:**

At least 99.9% availability, measured monthly. Container healthchecks and automatic restart policies support this target.

### **Measurement:**

UptimeRobot monitoring the application health endpoint, with Spring Boot Actuator health checks and Prometheus uptime metrics.

### **Example test**

A monitor checks the health endpoint every 5 minutes. Availability is reported as the percentage of successful checks over the monitoring period.

## **Recoverability**

In case of total system failure, the platform must be fully restorable within 2 hours using automated fallback mechanisms.

### **Quantification:**

- Time from total failure to the health endpoint reporting UP is 2 hours or less.
- No committed time entries or approved timesheets are lost (restored from the latest backup).

### *Measurement:*

The automated deployment workflow, scheduled database backups, and a timed recovery drill.

### *Example test:*

Destroy the running stack on a test environment, rebuild it through the automated deploy path, restore the latest database backup, and time how long it takes until the health endpoint reports UP. The test passes at 2 hours or less with all committed rows present.

## Accessibility

The system must be usable by people with disabilities, including keyboard-only and screen-reader users, because time logging and approvals are daily workflows for every employee.

### *Quantification:*

- Lighthouse accessibility score of at least 90 on the Login, Dashboard, Timesheets and Insights screens.
- 0 critical axe-core violations on the same four screens.

### *Measurement:*

Chrome Lighthouse and axe DevTools, with Angular Material and semantic HTML as the design tactics.

### *Example test:*

Log in as a seeded user, open each of the four screens, and run Lighthouse (Accessibility) and an axe full-page scan. The test passes if every score is 90 or above and axe reports no critical violations.

## Security

The system handles sensitive employee data (timesheets, productivity scores, personal details) and must protect it from unauthorised access, in line with POPIA.

### Quantification:

- 100% of API endpoints containing protected data require authentication, except explicitly public endpoints.
- 100% of authorisation decisions are enforced server-side using Spring Security.
- Administrative accounts must not be able to access administrative functionality without successfully completing MFA
- 0 critical or high-severity vulnerabilities found by an OWASP ZAP baseline scan.
- SonarQube Cloud security rating of A on [main](#) (no open security issues of Medium severity or higher).
- Security review rating of A (100% of security hotspots reviewed).
- 0 open critical or high Dependabot alerts before a production release.

### Measurement:

*SonarQube Cloud (static analysis and hotspot review) and GitHub Dependabot (dependency scanning), alongside Spring Security, JWT and ZAP.*

### Example test:

Call every protected endpoint with no token and confirm each returns 401 or 403, then run a ZAP baseline scan against the deployed application. The test passes if all calls are rejected and ZAP reports no High findings.

Open the SonarQube Cloud dashboard for [main](#) and confirm the security rating is A and the security review rating is A, and confirm the Dependabot tab shows no open critical or high alerts. The test passes if both hold.

## Maintainability

The system must be easy to extend, and test. As new features are added (additional external integrations, cost management, business owner features), the codebase must remain understandable and modifiable without breaking existing functionality. The architecture must support adding new external SSO providers or switching between them with minimal effort.

### Quantification:

- At least 80% automated test coverage for backend business logic
- At least 70% automated test coverage for frontend business logic

- Adding a new external integration must require changes primarily within the relevant integration adapter package
- At least 90% of existing automated tests must continue to pass after the addition of a new integration
- SonarQube Cloud maintainability rating of A.
- Duplication of 3% or less on new code.
- Coverage reported through SonarQube Cloud (JaCoCo and Icov) against the 80% backend and 70% frontend targets.
- The quality gate must pass on every pull request. This gives your line "new features must not introduce new critical or high-severity issues" an actual enforcement point.

#### Measurement:

- JUnit 5
- Mockito
- SpringBoot Test
- Jest
- JaCoCo for Java code coverage

#### Example test:

Architecture should enforce that business logic must not directly depend on:

- Jira SDKs
- Google Calender API implementation classes
- GitHub API implementation classes
- SSO provider specific implementation classes

Instead adding a new integration just involves creating a new adapter instead of rewriting core business logic.

Look into using ArchUnit to fail a build if architectural boundaries are violated.

## Mapping Quality Requirements to Architectural Decisions

| Quality Requirement                          | Architectural Decision(s)                                                                                                                                                                                                               | Trade-off                                                                                                                                                                                                                               |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Performance: 95% of request $\leq$ 2 seconds | <ul style="list-style-type: none"> <li>- Redis caching for frequently accessed data</li> <li>- Database indexing</li> <li>- Asynchronous processing for non critical operations</li> <li>- Optimized JPA queries using JOINS</li> </ul> | <ul style="list-style-type: none"> <li>- Caching introduces that stale data may be served</li> <li>- Indexing may slow down write operations</li> <li>- Users may not get immediate confirmation for non-critical operations</li> </ul> |

|                                                                  |                                                                                                                                                                                                                  |                                                                                                                                                                                                                    |
|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Performance: 100 concurrent users                                | Stateless services layer                                                                                                                                                                                         | Session data must be stored externally                                                                                                                                                                             |
| Scalability: 200% workload increase with $\leq 10\%$ degradation | <ul style="list-style-type: none"> <li>- Stateless services layer</li> <li>- Database read replicas</li> </ul>                                                                                                   | <ul style="list-style-type: none"> <li>- Replication lag causing inconsistent reads</li> <li>- Writes still go to primary, which can still cause a bottleneck</li> </ul>                                           |
| Security: 100% authorization required                            | <ul style="list-style-type: none"> <li>- Spring Security with JWT</li> <li>- RBAC enforcement</li> <li>- AWS Parameter Store</li> <li>- BCrypt password encoding</li> </ul>                                      | <ul style="list-style-type: none"> <li>- Security checks cause latency on requests</li> <li>- Extra network calls to retrieve secrets</li> </ul>                                                                   |
| Security: MFA enforcement                                        | <ul style="list-style-type: none"> <li>- MFA for admin accounts</li> </ul>                                                                                                                                       | <ul style="list-style-type: none"> <li>- MFA can cause user frustration</li> <li>- Increases log in time</li> </ul>                                                                                                |
| Availability $\geq 99.9\%$ uptime                                | <ul style="list-style-type: none"> <li>- Health monitoring (AWS cloud watch and Spring Boot Actuator health endpoints)</li> <li>- Auto restart on failure</li> <li>- Load balancer with health checks</li> </ul> | <ul style="list-style-type: none"> <li>- Health checks add overhead</li> <li>- Auto restart may mask underlying problems</li> </ul>                                                                                |
| Availability: Minimum 2hr recovery                               | <ul style="list-style-type: none"> <li>- Automated backups</li> <li>- Failover replication</li> </ul>                                                                                                            | <ul style="list-style-type: none"> <li>- Transactions still being processed may be lost (in-flight transactions)</li> </ul>                                                                                        |
| Availability: Minimum 1% data loss                               | <ul style="list-style-type: none"> <li>- ACID transactions</li> <li>- Foreign key constraints</li> </ul>                                                                                                         | <ul style="list-style-type: none"> <li>- ACID causes database contention and locks</li> <li>- Foreign keys make writes slower</li> </ul>                                                                           |
| Maintainability: 80% test coverage                               | <ul style="list-style-type: none"> <li>- JUnit 5 and Mockito for unit tests</li> <li>- Jest for frontend tests</li> <li>- JaCoco for coverage reporting</li> </ul>                                               | <ul style="list-style-type: none"> <li>- Testing add to CI pipeline build time</li> <li>- Time to write tests</li> <li>- Tests need to be updated as code changes</li> </ul>                                       |
| Maintainability: Isolated adapter changes                        | <ul style="list-style-type: none"> <li>- Layered architecture</li> <li>- Adapter pattern for integrations</li> <li>- ArchUnit for architectural enforcement</li> </ul>                                           | <ul style="list-style-type: none"> <li>- Abstraction increases code complexity</li> <li>- Adapter classes increase the number of components to maintain</li> <li>- ArchUnit creates build time overhead</li> </ul> |

\*ACID - Atomacity, Consistency, Isolation, Durability

Table 1: Mapping Quality Requirements to Architectural Decisions

# NRF Traceability Matrix

| ID  | Requirement                              | Test / tool                        | Target                  | Actual so far                                                                                                                                                                                                                                   |
|-----|------------------------------------------|------------------------------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P-1 | Normal load (25 users) responds in 1-2s  | k6, 25 users, 3 min                | p(95) ≤ 2s, errors < 1% | <b>Met</b> p(95) 1.34s, median 55ms, 0% errors                                                                                                                                                                                                  |
| S-1 | 100 concurrent users without degradation | k6, 100 users, 3 min               | p(95) ≤ 2s, errors < 1% | <b>Met</b> 0% errors across all runs. At 100 VUs, p(95) was 263.64ms for <a href="#">/api/projects</a> and 281.62ms for the unread-count, active-timer and insights endpoints, essentially unchanged from 20 VUs, so no degradation under load. |
| A-1 | 99.9% monthly availability               | UptimeRobot on the health endpoint | ≥ 99.9%                 | <b>Met</b> , 99.188%<br>Monitoring is ongoing; a full month is not yet observable, so this is an early reading, not a monthly figure.<br>.                                                                                                      |

|       |                                            |                                                                                        |                   |                                                            |
|-------|--------------------------------------------|----------------------------------------------------------------------------------------|-------------------|------------------------------------------------------------|
| R-1   | Full restore within 2h                     | Recovery drill: destroy, rebuild via automated path, restore backup, time to health UP | ≤ 2h, 0 rows lost |                                                            |
| ACC-1 | Lighthouse accessibility ≥ 90 on 4 screens | Chrome Lighthouse                                                                      | ≥ 90              | <b>Met</b><br>95.3 SCORE for accessibility across 5 pages. |
| ACC-2 | 0 critical axe violations on 4 screens     | axe DevTools                                                                           | 0 critical        | <b>Met</b><br>0 critical found                             |
| ACC-3 | Static accessibility checks                | SonarQube Cloud (accessibility tag)                                                    | 0 open            | <b>Met</b><br>0 Open                                       |
| SEC-1 | Protected endpoints require authentication | Call each protected endpoint with no token                                             | 401/403 on all    | <b>Met</b>                                                 |
| SEC-2 | 0 high/critical vulnerabilities            | OWASP ZAP baseline scan                                                                | 0 High            |                                                            |

|       |                                                  |                       |                   |          |
|-------|--------------------------------------------------|-----------------------|-------------------|----------|
| SEC-3 | Static analysis shows no medium+ security issues | SonarQube Cloud, main | Security rating A | Rating B |
|-------|--------------------------------------------------|-----------------------|-------------------|----------|

|       |                                             |                                              |                                     |                                                                                                              |
|-------|---------------------------------------------|----------------------------------------------|-------------------------------------|--------------------------------------------------------------------------------------------------------------|
| SEC-4 | Hotspots reviewed                           | SonarQube Cloud                              | Security review rating A (100%)     | Rating E, However upon further inspection, sonarqube does not take into consideration the existence of nginx |
| SEC-5 | No high/critical dependency vulnerabilities | GitHub Dependabot                            | 0 open high/critical                | 0 open                                                                                                       |
| M-1   | Quality gate passes on every PR             | SonarQube Cloud quality gate                 | Gate passed                         | <b>Met</b>                                                                                                   |
| M-2   | Duplication and maintainability             | SonarQube Cloud                              | Duplication ≤ 3%, maintainability A | <b>Met</b><br>Duplication 0.7% overall                                                                       |
| M-3   | Test coverage                               | JaCoCo and Jest, reported to SonarQube Cloud | ≥ 70% backend, ≥ 70% frontend       |                                                                                                              |

## Bottleneck analysis (S-1)

The first bottleneck at 100 users was the database connection pool. With the default HikariCP size of 10, requests queued waiting for a free connection. Raising it to 30 with `spring.datasource.hikari.maximum-pool-size` cut p(95) from 11.54s to 6.98s and removed the errors, but the 2 second target is still missed. The remaining suspected bottlenecks are unpaginated list endpoints (about 61 KB per response) and no caching of read-heavy data. Each fix is applied one at a time with a k6 rerun so the effect of each change is visible.

Root cause was an N+1 query pattern in `getMyTasks`, with a project lookup, access check and assignee lookup per task. **Fix: batched lookups**, with project names loaded in one `findAllById` query, access checked once per distinct project, and the assignee name resolved once



# EVIDENCE

```

Grafana

execution: local
script: load-test.js
output: -

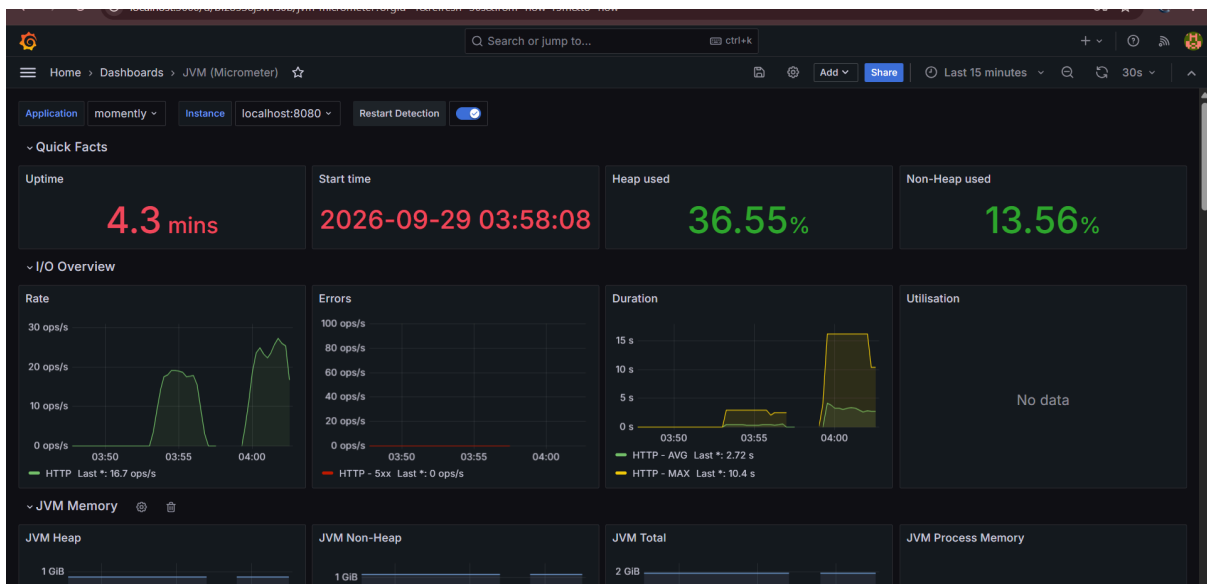
scenarios: (100.00%) 1 scenario, 25 max VUs, 3m30s max duration (incl. graceful stop):
  * default: 25 looping VUs for 3m0s (gracefulStop: 30s)

✓ GET /api/projects is 200
✓ GET /api/tasks/my-tasks is 200
✓ GET /api/timesheets/me is 200

[ ] setup
  ✓ login succeeded

checks.....: 100.00% 3289 out of 3289
data_received.....: 202 MB 1.1 MB/s
data_sent.....: 1.3 MB 7.0 MB/s
http_req_blocked.....: avg=25.48µs min=2.76µs med=9.96µs max=7.73ms p(90)=14.03µs p(95)=20.7µs
http_req_connecting.....: avg=11.72µs min=0s med=0s max=7.6ms p(90)=0s p(95)=0s
✓ http_req_duration.....: avg=385.37ms min=8.78ms med=55.15ms max=2.98s p(90)=1.14s p(95)=1.34s
  { expected_response:true }...: avg=385.37ms min=8.78ms med=55.15ms max=2.98s p(90)=1.14s p(95)=1.34s
✓ http_req_failed.....: 0.00% 0 out of 3289
http_req_receiving.....: avg=3.51ms min=24.18µs med=1.39ms max=60.43ms p(90)=9.13ms p(95)=12.42ms
http_req_sending.....: avg=49.84µs min=7.38µs med=29.36µs max=4.66ms p(90)=62.93µs p(95)=106.23µs
http_req_tls_handshaking.....: avg=0s min=0s med=0s max=0s p(90)=0s p(95)=0s
http_req_waiting.....: avg=381.8ms min=8.05ms med=53.38ms max=2.98s p(90)=1.13s p(95)=1.33s
http_reqs.....: 3289 17.799794/s
iteration_duration.....: avg=4.16s min=3.39s med=4.1s max=6.15s p(90)=4.76s p(95)=5.06s
iterations.....: 1096 5.031461/s
vus.....: 4 min=4 max=25
vus_max.....: 25 min=25 max=25

running (3m04.8s), 00/25 VUs, 1096 complete and 0 interrupted iterations
default / [=====] 25 VUs 3m0s
```



momently

Every moment counts

Dashboard

Timesheets

Log Time

Projects

My Tasks

Calendar

Leave Requests

Insights

Settings

EK

Enzokuhle Khumalo

enzokuhle.khumalo@momentum.co.za

Developer

Search projects...

PROJECTS

3

Total assigned projects

ACTIVE

2

Currently active

COMPLETED

0

Completed projects

YOUR HOURS

1650h

30m

Your hours across all projects

All

Active

On Hold

Completed

Archived

@ 2026 momently

Every Moment counts

5:20:10 AM - momentlyapp.c

https://momentlyapp.org/projects

96

Accessibility

These checks highlight opportunities to [improve the accessibility of your web app](#). Automatic detection can only detect a subset of issues and does not guarantee the accessibility of your web app, so [manual testing](#) is also encouraged.

CONTRAST

Background and foreground colors do not have a sufficient contrast ratio.

These are opportunities to improve the legibility of your content.

ADDITIONAL ITEMS TO MANUALLY CHECK (10)

These items address areas which an automated testing tool cannot cover. Learn more in our guide on [conducting an accessibility review](#).

PASSED AUDITS (23)

NOT APPLICABLE (39)

momently

Every moment counts

Dashboard

Timesheets

Log Time

Projects

My Tasks

Calendar

Leave Requests

Insights

Settings

EK

Enzokuhle Khumalo

enzokuhle.khumalo@momentum.co.za

Developer

Developer Insights

Last 4 weeks

Weekly summary

AI-generated overview of your recent activity

No summary generated yet for this week.

Generate summary

Your overview

Your productivity and development activity at a glance.

Productivity score

Your score for the selected period

83 / 100

↑ 3% vs last week

@ 2026 momently

Every Moment counts

5:19:41 AM - momentlyapp.c

https://momentlyapp.org/insights

96

Accessibility

These checks highlight opportunities to [improve the accessibility of your web app](#). Automatic detection can only detect a subset of issues and does not guarantee the accessibility of your web app, so [manual testing](#) is also encouraged.

CONTRAST

Background and foreground colors do not have a sufficient contrast ratio.

These are opportunities to improve the legibility of your content.

ADDITIONAL ITEMS TO MANUALLY CHECK (10)

These items address areas which an automated testing tool cannot cover. Learn more in our guide on [conducting an accessibility review](#).

PASSED AUDITS (23)

NOT APPLICABLE (39)

momently

Every moment counts

Dashboard

Timesheets

Log Time

Projects

My Tasks

Calendar

Leave Requests

Insights

Settings

EK

Enzokuhle Khumalo

enzokuhle.khumalo@momentum.co.za

Developer

Log Time

Log and manage your time entries.

Tue, Sep 29

Day

SUBMITTED

Submit Timesheet

The timesheet is awaiting review.

+ Add Manual Time Entry

Start Automatic Time Entry

Today's Entries

Total: 9h 30m

All

DRAFT

SUBMITTED

APPROVED

REJECTED

Project

All projects

From

09/29/2026

To

09/29/2026

| Time          | Project / Task         | Type   | Duration | Descrip                          |
|---------------|------------------------|--------|----------|----------------------------------|
| 09:00 - 18:30 | Mobile App Development | MANUAL | 9h 30m   | UI polis for dashb extend sessio |

@ 2026 momently

Every Moment counts

5:19:12 AM - momentlyapp.c

https://momentlyapp.org/log-time

91

Accessibility

These checks highlight opportunities to [improve the accessibility of your web app](#). Automatic detection can only detect a subset of issues and does not guarantee the accessibility of your web app, so [manual testing](#) is also encouraged.

NAMES AND LABELS

Select elements do not have associated label elements.

These are opportunities to improve the semantics of the controls in your application. This may enhance the experience for users of assistive technology, like a screen reader.

CONTRAST

Background and foreground colors do not have a sufficient contrast ratio.

These are opportunities to improve the legibility of your content.

ADDITIONAL ITEMS TO MANUALLY CHECK (10)

Show

These items address areas which an automated testing tool cannot cover. Learn more in our guide on [conducting an accessibility review](#).

momently

Every moment counts

Dashboard

Timesheets

Log Time

Projects

My Tasks

Calendar

Leave Requests

Insights

Settings

EK

Enzokuhle Khumalo

enzokuhle.khumalo@momentum.co.za

Developer

Timesheets

View and manage your timesheets.

All

Draft

Submitted

Approved

Rejected

| PERIOD                                                  | STATUS    |
|---------------------------------------------------------|-----------|
| <div>Week 41</div> <div>06 Oct 2025 - 12 Oct 2025</div> | SUBMITTED |
| <div>Week 42</div> <div>13 Oct 2025 - 19 Oct 2025</div> | SUBMITTED |
| <div>Week 43</div> <div>20 Oct 2025 - 26 Oct 2025</div> | SUBMITTED |
| <div>Week 44</div> <div>27 Oct 2025 - 02 Nov 2025</div> | SUBMITTED |
| <div>Week 45</div> <div>03 Nov 2025 - 09 Nov 2025</div> | SUBMITTED |
| <div>Week 46</div> <div>10 Nov 2025 - 16 Nov 2025</div> | SUBMITTED |

@ 2026 momently

Every Moment counts

5:18:48 AM - momentlyapp.c

https://momentlyapp.org/timesheets

100

Accessibility

These checks highlight opportunities to [improve the accessibility of your web app](#). Automatic detection can only detect a subset of issues and does not guarantee the accessibility of your web app, so [manual testing](#) is also encouraged.

ADDITIONAL ITEMS TO MANUALLY CHECK (10)

Hide

Interactive controls are keyboard focusable

Interactive elements indicate their purpose and state

The page has a logical tab order

Visual order on the page follows DOM order

User focus is not accidentally trapped in a region

The user's focus is directed to new content added to the page

HTML5 landmark elements are used to improve navigation

Offscreen content is hidden from assistive technology

13

```

Grafana
execution: local
script: load-test.js
output: -

scenarios: (100.00%) 2 scenarios, 100 max VUs, 5m0s max duration (incl. graceful stop):
  * normal: 20 looping VUs for 2m0s (exec: run, gracefulStop: 30s)
  * peak: 100 looping VUs for 2m0s (exec: run, startTime: 2m30s, gracefulStop: 30s)

✓ /api/notifications/unread/count 2xx
✓ /api/timers/active 2xx
✓ /api/insights/me 2xx

■ setup
  ✓ login succeeded

checks.....: 100.00% 11605 out of 11605
data_received.....: 6.4 MB 23 MB/s
data_sent.....: 509 KB 1.9 KB/s
http_req_blocked.....: avg=7.2ms min=135ns med=644ns max=1.8s p(90)=1.29µs p(95)=1.58µs
http_req_connecting.....: avg=2.14ms min=0s med=0s max=1.26s p(90)=0s p(95)=0s
http_req_duration.....: avg=247.25ms min=234.22ms med=239.51ms max=394.89ms p(90)=252.83ms p(95)=275.31ms
  { expected_response:true } : avg=247.25ms min=234.22ms med=239.51ms max=394.89ms p(90)=252.83ms p(95)=275.31ms
  { load:normal } : avg=241.26ms min=235.41ms med=238.94ms max=309.3ms p(90)=246.91ms p(95)=254.67ms
  { load:peak } : avg=248.44ms min=234.22ms med=239.65ms max=394.89ms p(90)=254.85ms p(95)=281.62ms
http_req_failed.....: 0.00% 0 out of 11605
http_req_receiving.....: avg=310.81µs min=13.54µs med=110.8µs max=142.55ms p(90)=189.49µs p(95)=283.8µs
http_req_sending.....: avg=148.97µs min=15.36µs med=107.7µs max=96.39ms p(90)=229.57µs p(95)=289.3µs
http_req_tls_handshaking.....: avg=5.05ms min=0s med=0s max=1.22s p(90)=0s p(95)=0s
http_req_waiting.....: avg=246.8ms min=234.01ms med=239.16ms max=323.28ms p(90)=252.18ms p(95)=274.35ms
http_reqs.....: 11605
iteration_duration.....: avg=3.76s min=3.7s med=3.72s max=0.52s p(90)=3.8s p(95)=4.01s
iterations.....: 3868 10.099217/s
vus.....: 9 min=0 max=100
vus_max.....: 100 min=0 max=100

running (4m35.1s), 000/100 VUs, 3868 complete and 0 interrupted iterations
normal / [=====] 20 VUs 2m0s
peak / [=====] 100 VUs 2m0s

```

UptimeRobot

Monitoring

3rd party services

Incidents

Status pages

Maintenance

Team members

Integrations & API

Cybernauts...

Upgrade now

Monitoring

momentlyapp.org

HTTP/S monitor for <https://momentlyapp.org/>

Test Notification

Pause

Edit

Current status

Up

Currently up for 18h 20m 28s

Last check

1m, 19s ago

Checked every 5m [Get 60 sec. checks](#)

Last 24 hours

98.775%

2 incidents, 15m, 53s down

Domain & SSL

Domain valid until

Unlock

SSL certificate valid until

Unlock

Available only in Solo, Team and Scale [Upgrade now](#)

Next maintenance

No maintenance planned.

Set up maintenance

Regions

Last 7 days

98.775%

2 incidents, 15m, 53s down

Last 30 days

98.775%

2 incidents, 15m, 53s down

Last 365 days

--.---%

Unlock with paid plans

Pick a dat...

MTBF 7 days

10.68 hours

Response time for All regions

Setup alerts For slow response times

Last hour

97ms

50ms

25ms

sonarqubecloud Bot commented 2 hours ago

Quality Gate passed

Issues

0 New issues

0 Accepted issues

Measures

0 Security Hotspots

0.0% Coverage on New Code

0.0% Duplication on New Code

See analysis details on SonarQube Cloud

## Maintainability snapshot

### Maintainability rating

Overall code

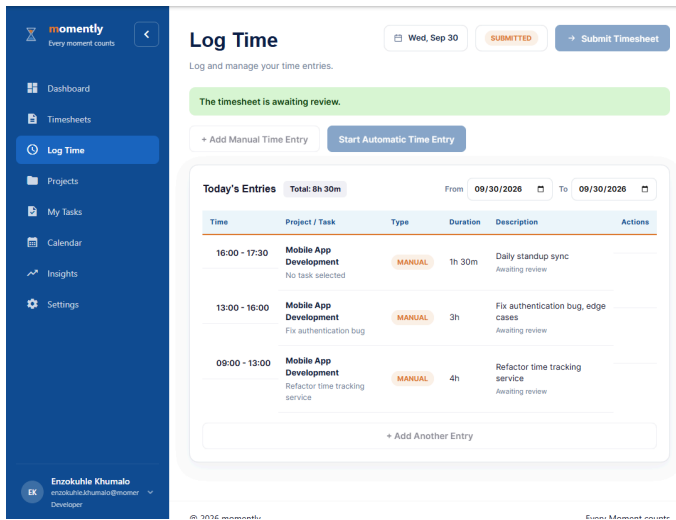
A

## Reliability snapshot

### Reliability rating

Overall code

A



**Log Time**  
Log and manage your time entries.

The timesheet is awaiting review.

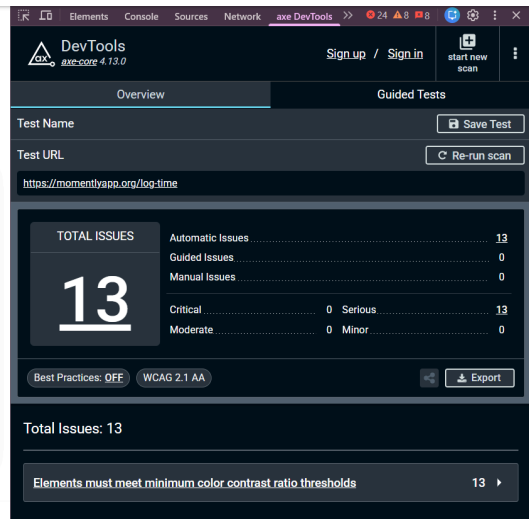
+ Add Manual Time Entry   Start Automatic Time Entry

**Today's Entries**   Total: 8h 30m   From 09/30/2026   To 09/30/2026

| Time          | Project / Task                                           | Type   | Duration | Description                                           | Actions |
|---------------|----------------------------------------------------------|--------|----------|-------------------------------------------------------|---------|
| 16:00 - 17:30 | Mobile App Development<br>No task selected               | MANUAL | 1h 30m   | Daily standup sync<br>Awaiting review                 |         |
| 13:00 - 16:00 | Mobile App Development<br>Fix authentication bug         | MANUAL | 3h       | Fix authentication bug, edge cases<br>Awaiting review |         |
| 09:00 - 13:00 | Mobile App Development<br>Refactor time tracking service | MANUAL | 4h       | Refactor time tracking service<br>Awaiting review     |         |

+ Add Another Entry

Enzokuhle Khumalo  
enzokuhle.khumalo@momently.com  
Developer



**axe DevTools**  
axe-core 4.13.0

Sign up / Sign in   start new scan

Overview   Guided Tests

Test Name   Save Test

Test URL   Re-run scan

https://momentlyapp.org/log-time

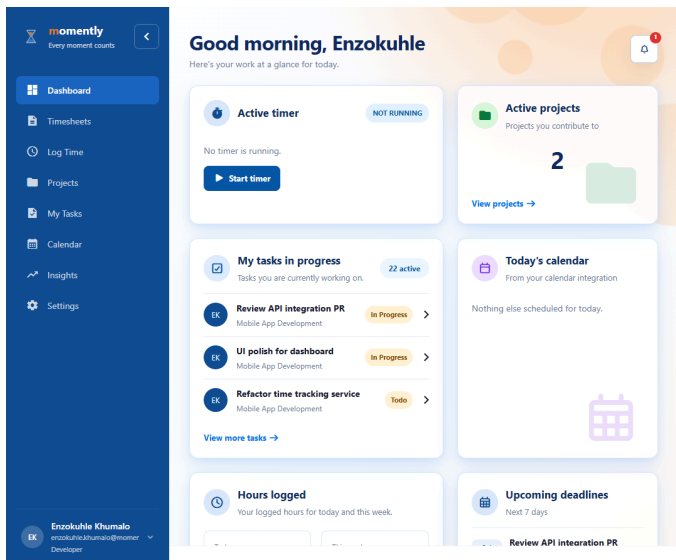
**TOTAL ISSUES**  
13

| Issue Type       | Count |
|------------------|-------|
| Automatic Issues | 13    |
| Guided Issues    | 0     |
| Manual Issues    | 0     |
| Critical         | 0     |
| Serious          | 13    |
| Moderate         | 0     |
| Minor            | 0     |

Best Practices: OFF   WCAG 2.1 AA   Export

Total Issues: 13

Elements must meet minimum color contrast ratio thresholds   13



**Good morning, Enzokuhle**  
Here's your work at a glance for today.

**Active timer**   NOT RUNNING  
No timer is running.   Start timer

**Active projects**  
Projects you contribute to  
2   View projects →

**My tasks in progress**   22 active  
Tasks you are currently working on.

- Review API integration PR   Mobile App Development   In Progress
- UI polish for dashboard   Mobile App Development   In Progress
- Refactor time tracking service   Mobile App Development   Todo

View more tasks →

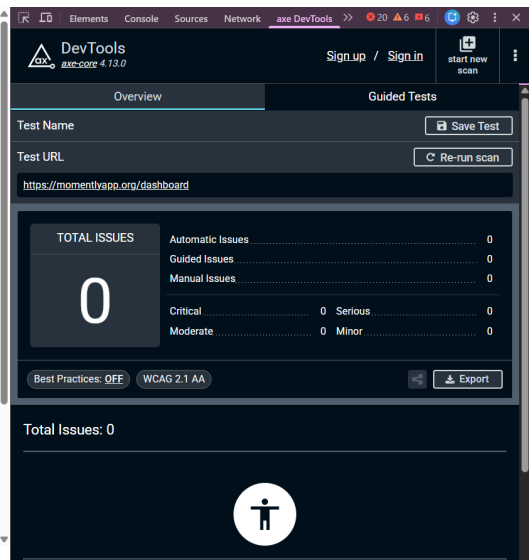
**Today's calendar**  
From your calendar integration  
Nothing else scheduled for today.

**Hours logged**  
Your logged hours for today and this week.

**Upcoming deadlines**  
Next 7 days

Review API integration PR

Enzokuhle Khumalo  
enzokuhle.khumalo@momently.com  
Developer



**axe DevTools**  
axe-core 4.13.0

Sign up / Sign in   start new scan

Overview   Guided Tests

Test Name   Save Test

Test URL   Re-run scan

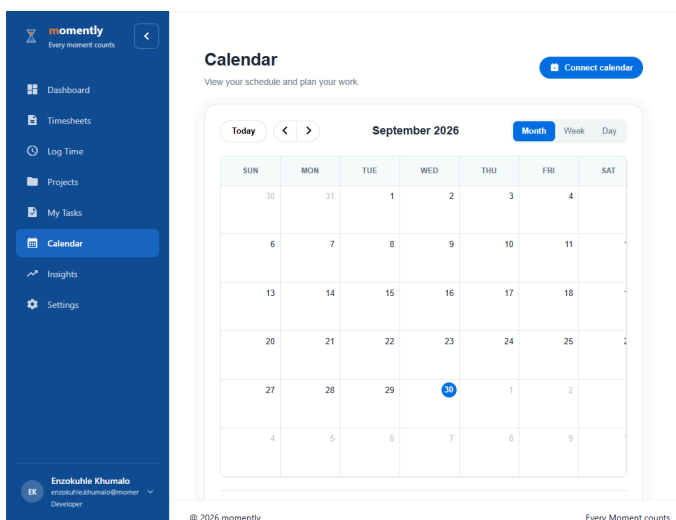
https://momentlyapp.org/dashboard

**TOTAL ISSUES**  
0

| Issue Type       | Count |
|------------------|-------|
| Automatic Issues | 0     |
| Guided Issues    | 0     |
| Manual Issues    | 0     |
| Critical         | 0     |
| Serious          | 0     |
| Moderate         | 0     |
| Minor            | 0     |

Best Practices: OFF   WCAG 2.1 AA   Export

Total Issues: 0



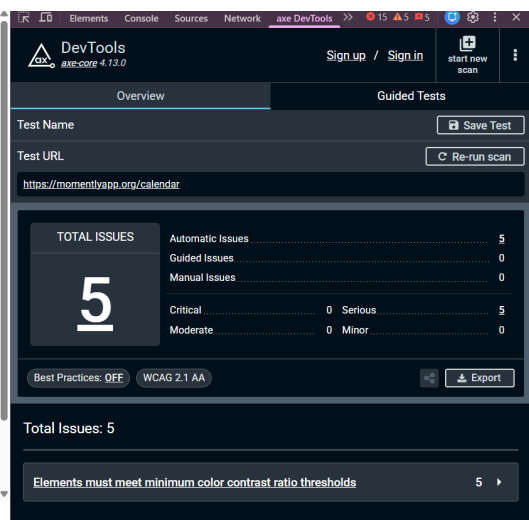
**Calendar**  
View your schedule and plan your work.

Connect calendar

Today   September 2026   Month   Week   Day

| SUN | MON | TUE | WED | THU | FRI | SAT |
|-----|-----|-----|-----|-----|-----|-----|
| 30  | 31  | 1   | 2   | 3   | 4   |     |
| 6   | 7   | 8   | 9   | 10  | 11  |     |
| 13  | 14  | 15  | 16  | 17  | 18  |     |
| 20  | 21  | 22  | 23  | 24  | 25  |     |
| 27  | 28  | 29  | 30  | 1   | 2   |     |
| 4   | 5   | 6   | 7   | 8   | 9   |     |

Enzokuhle Khumalo  
enzokuhle.khumalo@momently.com  
Developer



**axe DevTools**  
axe-core 4.13.0

Sign up / Sign in   start new scan

Overview   Guided Tests

Test Name   Save Test

Test URL   Re-run scan

https://momentlyapp.org/calendar

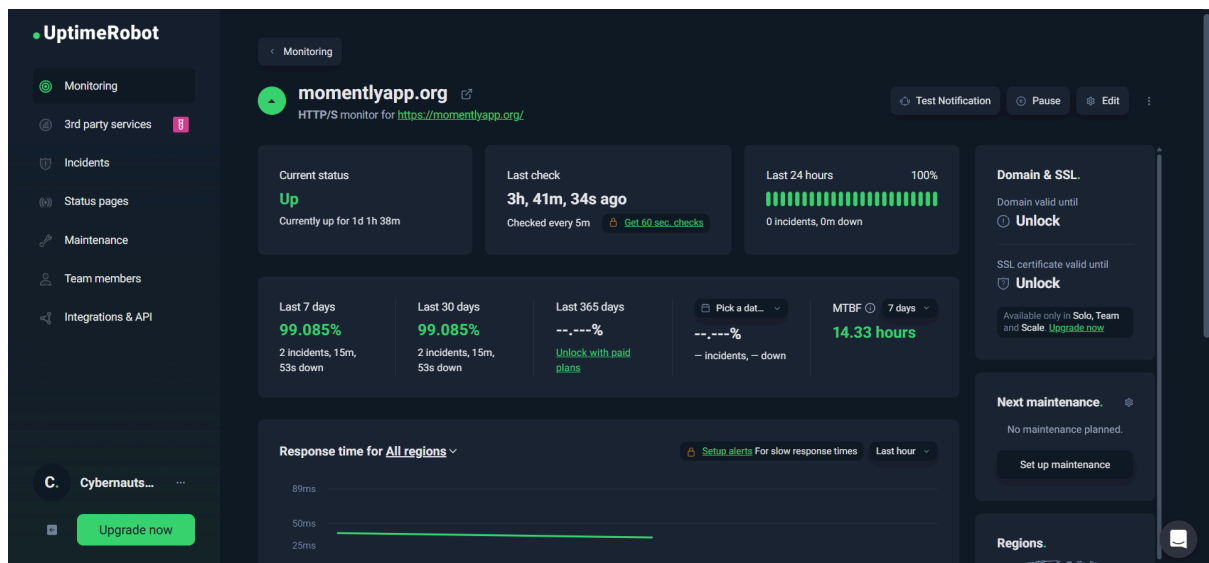
**TOTAL ISSUES**  
5

| Issue Type       | Count |
|------------------|-------|
| Automatic Issues | 5     |
| Guided Issues    | 0     |
| Manual Issues    | 0     |
| Critical         | 0     |
| Serious          | 5     |
| Moderate         | 0     |
| Minor            | 0     |

Best Practices: OFF   WCAG 2.1 AA   Export

Total Issues: 5

Elements must meet minimum color contrast ratio thresholds   5



## TESTING THE '/'

10–25 VUs → baseline load

```
stages: [
  { duration: '30s', target: 10},
  { duration: '1m', target: 10},
  { duration: '30s', target: 25},
  { duration: '1m', target: 25},
  { duration: '30s', target: 0},
]
```

p(95) < 250 ms    235.79 ms  
 p(99) < 1000 ms    432.52 ms  
 failure rate < 1%    0.00%

**Baseline Load Test:** The system was tested with up to 25 concurrent virtual users over approximately 3.5 minutes. The test generated 14 340 HTTP requests at an average throughput of 62.7 requests/second.

No HTTP request failures occurred. The 95th-percentile response time was 235.79 ms and the 99th-percentile response time was 432.52 ms. During observation, container resource utilisation remained low, with the backend using approximately 295 MiB of memory and 0.20% CPU at the sampled point.

## 50 VUs

```
stages: [
  { duration: '30s', target: 10},
  { duration: '1m', target: 10},
  { duration: '30s', target: 50},
  { duration: '1m', target: 50},
  { duration: '30s', target: 0},
]
```

| Metric       | 25 VUs     | 50 VUs      | Interpretation          |
|--------------|------------|-------------|-------------------------|
| Throughput   | 62.7 req/s | 107.2 req/s | Increased substantially |
| Average      | 237.72 ms  | 238.16 ms   | Essentially unchanged   |
| p95          | 235.79 ms  | 229.76 ms   | Essentially unchanged   |
| p99          | 432.52 ms  | 466.26 ms   | Small increase          |
| Failure rate | 0%         | 0%          | Excellent               |
| Max          | 2.94 s     | 2.95 s      | Essentially unchanged   |

p95 < 250 ms    229.76 ms  
 p99 < 1000 ms    466.26 ms  
 failure < 1%    0%

The system successfully supported 50 concurrent virtual users, processing 24 548 requests at an average throughput of 107.2 requests/second. No HTTP request failures were observed. Response-time performance remained stable, with an average latency of 238.16 ms, p95 latency of 229.76 ms, and p99 latency of 466.26 ms. The results indicate that the single-instance deployment comfortably handled the tested 50-VU workload without significant performance degradation.

## 100 VUs

| Metric     | 50 VUs      | 100 VUs     | Interpretation        |
|------------|-------------|-------------|-----------------------|
| Requests   | 24,548      | 37,240      | Higher                |
| Throughput | 107.2 req/s | 142.6 req/s | Higher                |
| Average    | 238.16 ms   | 243.13 ms   | Slight increase       |
| Median     | 217.33 ms   | 217.25 ms   | Essentially unchanged |

|              |           |                                 |  |
|--------------|-----------|---------------------------------|--|
| p95          | 229.76 ms | 237.80 ms   Slight increase     |  |
| p99          | 466.26 ms | 640.63 ms   Noticeable increase |  |
| Failure rate | 0%        | 0%   Same                       |  |
| Maximum      | 2.95 s    | 19.34 s   Found outlier         |  |

p95 < 250 ms  
p99 < 1000 ms  
failure < 1%

25 → 50 → 100 VUs did not cause significant degradation in p95 latency.

---

Testing using GET /api/timers/active

Full Endpoint: GET <https://momentlyapp.org/api/timers/active>